Q1.) What are different protocols you observe at the following layers of the protocol stack?

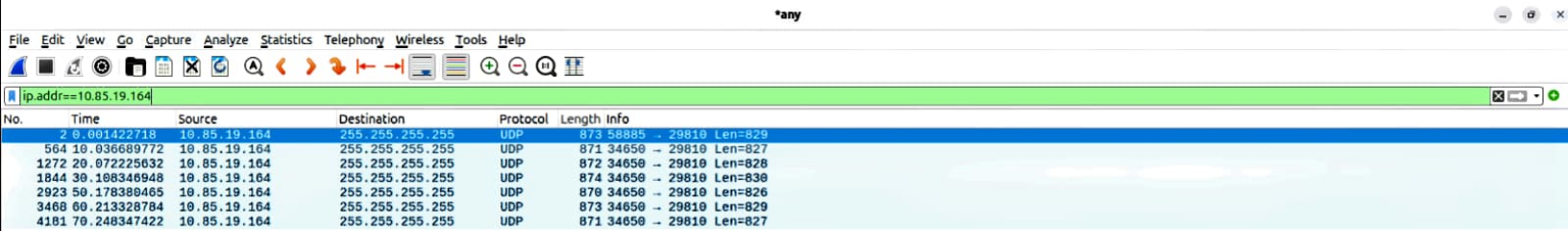
Ans ⇒

Application Layer	Transport Layer	Network Layer
• MDNS	• UDP	• ARP
• SSDP	• TCP	• ICMPv6
• DNS		
• NBNS		
• HTTP		
• DHCP		

Q2.) What is the total amount of data being received for the following two cases?

A.) when you access :- http://neverssl.com/

B.) when you access :- https://www.cornell.edu/


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr==10.85.19.164
No. Time Source Destination Protocol Length Info
2 0.001422718 10.85.19.164 255.255.255.255 UDP 873 58885 → 29810 Len=829
564 10.036689772 10.85.19.164 255.255.255.255 UDP 871 34650 → 29810 Len=827
1272 20.072225632 10.85.19.164 255.255.255.255 UDP 872 34650 → 29810 Len=828
1844 30.108346948 10.85.19.164 255.255.255.255 UDP 874 34650 → 29810 Len=830
2923 50.178380465 10.85.19.164 255.255.255.255 UDP 870 34650 → 29810 Len=826
3468 60.213328784 10.85.19.164 255.255.255.255 UDP 873 34650 → 29810 Len=829
4181 70.248347422 10.85.19.164 255.255.255.255 UDP 871 34650 → 29810 Len=827


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.addr==10.85.23.218
No. Time Source Destination Protocol Length Info
10 0.003916999 10.85.23.218 224.0.0.251 MDNS 202 Standard query 0x0049 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googleca…
1022 16.282850671 10.85.23.218 224.0.0.251 MDNS 202 Standard query 0x004a PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googleca…
2104 36.453191138 10.85.23.218 224.0.0.251 MDNS 202 Standard query 0x004b PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googleca…

B.) When you access :- https://www.cornell.edu/

Q3.) Answer the following when you access the site
http://neverssl.com/

A.) How many HTTP ~~Get~~ GET request do you observe? List down the GET requests.

2 HTT GET requests observed

GET requests :-

Time	source	Destination	Protocol	Length	Info
3.074771527	10.85.20.52	185.125.190.100	HTTP	156	GET/HTTP/1.1
36.658437557	10.85.20.52	34.223.124.95	HTTP	143	GET/HTTP/1.1


*any
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method=="GET"
No. Time Source Destination Protocol Length Info
239 3.074771527 10.85.20.52 185.125.190.100 HTTP 156 GET / HTTP/1.1
2117 36.658437557 10.85.20.52 34.223.124.45 HTTP 143 GET / HTTP/1.1

C.) For each of the HTTP GET request, find out:

(i) The total number of TCP segments being recieved.

1 Segment of TCP protocol

(ii) The total amount of data being received in the corresponding HTTP response message.

4261 bytes

```
▸ Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.20.52
▾ Transmission Control Protocol, Src Port: 80, Dst Port: 44788, Seq: 1, Ack: 76, Len: 4261
    Source Port: 80
    Destination Port: 44788
    [Stream index: 0]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 4261]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1745965817
    [Next Sequence Number: 4262    (relative sequence number)]
    Acknowledgment Number: 76    (relative ack number)
    Acknowledgment number (raw): 1513798250
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0xce60 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
    TCP payload (4261 bytes)
```

```
0020  0a 55 14 34 00 50 ae f4  68 11 52 f9 5a 3a ba 6a   ·U·4·P·· h·R·Z:·j
0030  80 18 00 e5 ce 60 00 00  01 01 08 0a c8 f5 3d 1c   ·····`·· ......=·
0040  69 eb 26 00 48 54 54 50  2f 31 2e 31 20 32 30 30   i·&·HTTP /1.1 200
0050  20 4f 4b 0d 0a 44 61 74  65 3a 20 57 65 64 2c 20    OK··Dat e: Wed,
0060  30 38 20 4a 75 6c 20 32  30 32 36 20 31 37 3a 32   08 Jul 2 026 17:2
0070  34 3a 34 31 20 47 4d 54  0d 0a 53 65 72 76 65 72   4:41 GMT ·· Server
0080  3a 20 41 70 61 63 68 65  2f 32 2e 34 2e 36 36 20   : Apache /2.4.66
0090  28 29 0d 0a 55 70 67 72  61 64 65 3a 20 68 32 2c   () ·· Upgr ade: h2,
00a0  68 32 63 0d 0a 43 6f 6e  6e 65 63 74 69 6f 6e 3a   h2c ·· Con nection:
00b0  20 55 70 67 72 61 64 65  0d 0a 4c 61 73 74 2d 4d    Upgrade ·· Last-M
00c0  6f 64 69 66 69 65 64 3a  20 57 65 64 2c 20 32 39   odified: · Wed, 29
00d0  20 4a 75 6e 20 32 30 32  32 20 30 30 3a 32 33 3a    Jun 202 2 00:23:
00e0  33 33 20 47 4d 54 0d 0a  45 54 61 67 3a 20 22 66   33 GMT ·· ETag: "f
00f0  37 39 2d 35 65 32 38 62  32 39 64 33 38 65 39 33   79-5e28b 29d38e93
0100  22 0d 0a 41 63 63 65 70  74 2d 52 61 6e 67 65 73   "·· Accep t-Ranges
0110  3a 20 62 79 74 65 73 0d  0a 43 6f 6e 74 65 6e 74   : bytes· ·Content
0120  2d 4c 65 6e 67 74 68 3a  20 33 39 36 31 0d 0a 56   -Length: · 3961 ··V
0130  61 72 79 3a 20 41 63 63  65 70 74 2d 45 6e 63 6f   ary: Acc ept-Enco
0140  64 69 6e 67 0d 0a 43 6f  6e 74 65 6e 74 2d 54 79   ding ·· Co ntent-Ty
0150  70 65 3a 20 74 65 78 74  2f 68 74 6d 6c 3b 20 63   pe: text /html; c
0160  68 61 72 73 65 74 3d 55  54 46 2d 38 0d 0a 0d 0a   harset=U TF-8····
0170  3c 68 74 6d 6c 3e 0a 09  3c 68 65 61 64 3e 0a 09   <html>·· <head>··
0180  09 3c 74 69 74 6c 65 3e  4e 65 76 65 72 53 53 4c    <title> NeverSSL
0190  20 2d 20 43 6f 6e 6e 65  63 74 69 6e 67 20 2e 2e    - Conne cting ..
01a0  2e 20 3c 2f 74 69 74 6c  65 3e 0a 09 09 3c 73 74   . </titl e>···<st
01b0  79 6c 65 3e 0a 09 09 62  6f 64 79 20 7b 0a 09 09   yle>··· b ody {···
01c0  09 66 6f 6e 74 2d 66 61  6d 69 6c 79 3a 20 4d 6f    font-fa mily: Mo
01d0  6e 74 73 65 72 72 61 74  2c 20 68 65 6c 76 65 74   ntserrat , helvet
```

TCP Segment Len (tcp.len)

✓ Show packet bytes

Help

(D.) Identify the TCP three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

① SYN PACKET

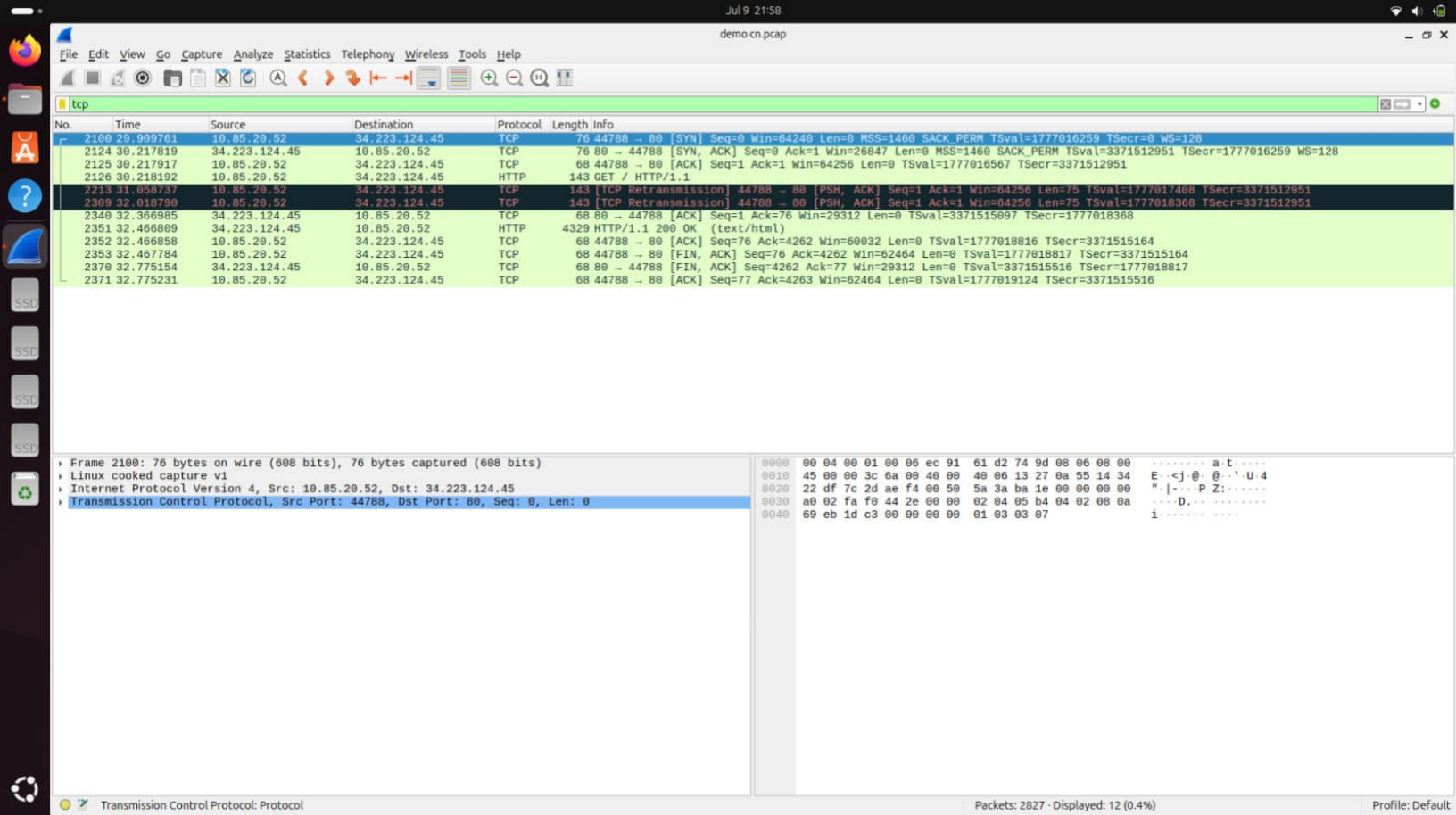
- Packet NO. : 217
- SOURCE IP : 10.05.20.05
- Destination IP : 105.125.190.100
- Sequence NO.(seq) : 0

② SYN-ACK PACKET

- Packet NO. : 237
- Source IP : 105.125.190.100
- Destination IP : 10.05.20.52
- Sequence Number (seq) : 0
  (Acknowlegment Number ACK = 1)

③ ACK Packet

- Packet NO. : 238
- Source IP : 10.05.20.52
- Destination IP : 105.125.190.100
- Sequence Number (seq) : 1


Jul 9 21:58
demo cn.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
2100 29.909761 10.85.20.52 34.223.124.45 TCP 76 44788 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1777016259 TSecr=0 WS=128
2124 30.217819 34.223.124.45 10.85.20.52 TCP 76 80 → 44788 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3371512951 TSecr=1777016259 WS=128
2125 30.217917 10.85.20.52 34.223.124.45 TCP 68 44788 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1777016567 TSecr=3371512951
2126 30.218192 10.85.20.52 34.223.124.45 HTTP 143 GET / HTTP/1.1
2213 31.058737 10.85.20.52 34.223.124.45 TCP 143 [TCP Retransmission] 44788 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=75 TSval=1777017408 TSecr=3371512951
2309 32.018790 10.85.20.52 34.223.124.45 TCP 143 [TCP Retransmission] 44788 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=75 TSval=1777018368 TSecr=3371512951
2340 32.366985 34.223.124.45 10.85.20.52 TCP 68 80 → 44788 [ACK] Seq=1 Ack=76 Win=29312 Len=0 TSval=3371515097 TSecr=1777018368
2351 32.466809 34.223.124.45 10.85.20.52 HTTP 4329 HTTP/1.1 200 OK (text/html)
2352 32.466858 10.85.20.52 34.223.124.45 TCP 68 44788 → 80 [ACK] Seq=76 Ack=4262 Win=60032 Len=0 TSval=1777018816 TSecr=3371515164
2353 32.467784 10.85.20.52 34.223.124.45 TCP 68 44788 → 80 [FIN, ACK] Seq=76 Ack=4262 Win=62464 Len=0 TSval=1777018817 TSecr=3371515164
2370 32.775154 34.223.124.45 10.85.20.52 TCP 68 80 → 44788 [FIN, ACK] Seq=4262 Ack=77 Win=29312 Len=0 TSval=3371515516 TSecr=1777018817
2371 32.775231 10.85.20.52 34.223.124.45 TCP 68 44788 → 80 [ACK] Seq=77 Ack=4263 Win=62464 Len=0 TSval=1777019124 TSecr=3371515516
Frame 2100: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
Linux cooked capture v1
Internet Protocol Version 4, Src: 10.85.20.52, Dst: 34.223.124.45
Transmission Control Protocol, Src Port: 44788, Dst Port: 80, Seq: 0, Len: 0
0000 00 04 00 01 00 06 ec 91 61 d2 74 9d 08 06 08 00
0010 45 00 00 3c 6a 00 40 00 40 06 13 27 0a 55 14 34
0020 22 df 7c 2d ae f4 00 50 5a 3a ba 1e 00 00 00 00
0030 a0 02 fa f0 44 2e 00 00 02 04 05 b4 04 02 08 0a
0040 69 eb 1d c3 00 00 00 00 01 03 03 07
Transmission Control Protocol: Protocol
Packets: 2827 · Displayed: 12 (0.4%)
Profile: Default